

Zachary M. Labe

CONTACT INFORMATION

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RESEARCH INTERESTS

large-scale climate variability • teleconnections • explainable AI • Arctic sea ice • data science methods • large ensembles • decadal prediction • extremes

EDUCATION

University of California, Irvine, Irvine, CA
Ph.D. in Earth System Science, June 2020

University of California, Irvine, Irvine, CA
M.Sc. in Earth System Science, December 2017

Cornell University, Ithaca, NY
B.Sc. in Atmospheric Science, May 2015, *Distinction in Research*
Minor: Dyson Business Minor for Life Sciences

RESEARCH EMPLOYMENT

2025-Present	Climate Scientist Climate Central, Inc.
2024-2025	Research Physical Scientist (NOAA Federal) Geophysical Fluid Dynamics Laboratory (GFDL), Seasonal-to-Decadal Variability and Predictability Division
2024	Associate Research Scholar Princeton University AOS/CIMES and NOAA GFDL
2022-2024	Postdoctoral Research Associate Advisors: Drs. Thomas Delworth and Nathaniel Johnson, Princeton University AOS/CIMES and NOAA GFDL
2020-2022	Postdoctoral Researcher Advisor: Dr. Elizabeth Barnes, Atmospheric Science, Colorado State University
2015-2020	Graduate Research Assistant Advisor: Dr. Gudrun Magnusdottir, Earth System Science, University of California, Irvine
2014-2015	Undergraduate Research Assistant Advisor: Dr. Toby Ault, Earth and Atmospheric Science, Cornell University

PUBLICATIONS

Refereed/Peer-Reviewed

- [40] Zhang, Y. and **Z.M. Labe** (2026). Adapting Infrastructure to a Changing Climate with Extreme Event Attribution Assessments. *Climate-Resilient Structures and Infrastructures*, DOI:10.1201/9781003567332-6. [\[Article\]](#)
- [39] Jong, B-T., **Z.M. Labe**, T.L. Delworth, and W.F. Cooke (2026). Reversal of extreme precipitation trends over the Northeast US in response to aggressive climate mitigation in GFDL SPEAR, *Environmental Research Letters*, DOI:10.1088/1748-9326/ae51a6. [\[Article\]](#)
- [38] Joh, Y., S-W. Yeh, T.L. Delworth, **Z.M. Labe**, A.T. Wittenberg, W.F. Cooke, J. Lou, and Y-G. Park (2026). Evolving synchronization of Gulf Stream and Kuroshio-Oyashio Extension in a changing climate. *Science Advances*, DOI: 10.1126/sciadv.adx6366. [\[Article\]](#)
- [37] Timmermans, M.-L. and **Z.M. Labe** (2025). Sea surface temperature [in "Arctic Report Card 2025"], NOAA, DOI:10.25923/pz7y-3b10. [\[Article\]](#)
- [36] Eayrs, C. and **Z.M. Labe** (2025). The future of sea ice. *Comprehensive Cryospheric Science and Environmental Change*, DOI:10.1016/B978-0-323-85242-5.00050-6. [\[Chapter\]](#)
- [35] Kalashnikov, D.A., J.T. Abatzoglou, F.V. Davenport, **Z.M. Labe**, P.C. Loikith, D. Touma, and D. Singh (2025). Projections of Lightning-Ignited Wildfire Risk in the Western United States. *Earth's Future*, DOI:10.1029/2025EF006108. [\[Article\]](#)
- [34] Timmermans, M.-L. and **Z.M. Labe** (2025). [The Arctic] Sea surface temperature [in "State of the Climate in 2024"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-25-0104.1. [\[Article\]](#)
- [33] Timmermans, M.-L. and **Z.M. Labe** (2024). Sea surface temperature [in "Arctic Report Card 2024"], NOAA, DOI:10.25923/9z96-aq19. [\[Article\]](#)
- [32] Kalashnikov, D.A., F.V. Davenport, **Z.M. Labe**, P.C. Loikith, J.T. Abatzoglou, and D. Singh (2024). Predicting cloud-to-ground lightning in the Western United States from the large-scale environment using explainable neural networks. *Journal of Geophysical Research: Atmospheres*, DOI:10.1029/2024JD042147. [\[Article\]](#)
- [31] Vanek S.B., **Z.M. Labe**, O. Lauter, K. Shionalyn, M.A. Shadab, E. Adasheva, A. Margevich, M.N. Schaberg, L. Ashokkumar, and J.N. Naoukin (2024). Exploring the past, present, and future of USAPECS: Lessons from a decade of supporting early career research across national and international polar networks. *Arctic Yearbook*, ISSN 2298-2418. [\[Article\]](#)
- [30] **Labe, Z.M.**, T.L. Delworth, N.C. Johnson, and W.F. Cooke (2024). Exploring a data-driven approach to identify regions of change associated with future climate scenarios. *Journal of Geophysical Research: Machine Learning and Computation*, DOI:10.1029/2024JH000327. [\[Article\]](#)
- [29] Schreck III, C.M., D.R. Easterling, J.J. Barsugli, D.A. Coates, A. Hoell, N.C. Johnson, K.E. Kunkel, **Z.M. Labe**, J. Uehling, R.S. Vose, and X. Zhang (2024). A rapid response process for evaluating causes of extreme temperature events in the United States: the 2023 Texas/Louisiana heat wave as a prototype. *Environmental Research: Climate*, DOI:10.1088/2752-5295/ad8028. [\[Article\]](#)
- [28] Kretschmer, M., A. Jézéquel, **Z.M. Labe**, and D. Touma (2024). A shifting climate: new paradigms and challenges for (early career) scientists in extreme weather research. *Atmospheric Science Letters*, DOI:10.1002/asl.1268. [\[Article\]](#)
- [27] Timmermans, M.-L. and **Z.M. Labe** (2024). [The Arctic] Sea surface temperature [in "State of the Climate in 2023"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-24-0101.1. [\[Article\]](#)
- [26] Bushuk, M., S. Ali, D. Bailey, Q. Bao, L. Batte, U.S. Bhatt, E. Blanchard-Wrigglesworth, E. Blockley, G. Cawley, J. Chi, F. Counillon, P. Goulet Coulombe, R. Cullather, F.X.

- Diebold, A. Dirkson, E. Exarchou, M. Gobel, W. Gregory, V. Guemas, L. Hamilton, B. He, S. Horvath, M. Ionita, J. E. Kay, E. Kim, N. Kimura, D. Kondrashov, **Z.M. Labe**, W. Lee, Y.J. Lee, C. Li, X. Li, Y. Lin, Y. Liu, W. Maslowski, F. Massonnet, W.N. Meier, W.J. Merryfield, H. Myint, J.C. Acosta Navarro, A. Petty, F. Qiao, D. Schroder, A. Schweiger, Q. Shu, M. Sigmond, M. Steele, J. Stroeve, N. Sun, S. Tietsche, M. Tsamados, K. Wang, J. Wang, W. Wang, Y. Wang, Y. Wang, J. Williams, Q. Yang, X. Yuan, J. Zhang, and Y. Zhang (2024). Predicting September Arctic sea ice: A multi-model seasonal skill comparison. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-23-0163.1. [\[Article\]](#)
- [25] Zhang, Y., B.M. Ayyub, J.F. Fung, and **Z.M. Labe** (2024). Incorporating extreme event attribution into climate change adaptation for civil infrastructure: Methods, benefits, and research needs. *Resilient Cities and Structures*, DOI:10.1016/j.rcns.2024.03.002. [\[Article\]](#)
- [24] **Labe, Z.M.**, N.C. Johnson, and T.L. Delworth (2024). Changes in United States summer temperatures revealed by explainable neural networks. *Earth's Future*, DOI:10.1029/2023EF003981. [\[Article\]](#) [\[GFDL Bulletin: Research Highlights\]](#)
- [23] Timmermans, M.-L. and **Z.M. Labe** (2023). Sea surface temperature [in "Arctic Report Card 2023"], NOAA, DOI:10.25923/e8jc-f342. [\[Article\]](#)
- [22] Timmermans, M.-L. and **Z.M. Labe** (2023). [The Arctic] Sea surface temperature [in "State of the Climate in 2022"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-23-0079.1. [\[Article\]](#)
- [21] Eischeid, J.K., M.P. Hoerling, X.-W. Quan, A. Kumar, J. Barsugli, **Z.M. Labe**, K.E. Kunkel, C.J. Schreck III, D.R. Easterling, T. Zhang, J. Uehling, and X. Zhang (2023). Why has the summertime central U.S. warming hole not disappeared? *Journal of Climate*, DOI:10.1175/JCLI-D-22-0716.1. [\[Article\]](#)
- [20] Witt, J.K., **Z.M. Labe**, A.C. Warden, and B.A. Clegg (2023). Visualizing Uncertainty in Hurricane Forecasts with Animated Risk Trajectories. *Weather, Climate, and Society*, DOI:10.1175/WCAS-D-21-0173.1. [\[Article\]](#)
- [19] **Labe, Z.M.**, E.A. Barnes, and J.W. Hurrell (2023). Identifying the regional emergence of climate patterns in the ARISE-SAI-1.5 simulations. *Environmental Research Letters*, DOI:10.1088/1748-9326/acc81a. [\[Article\]](#)
- [18] Timmermans, M.-L. and **Z.M. Labe** (2022). Sea surface temperature [in "Arctic Report Card 2022"], NOAA, DOI:10.25923/p493-2548. [\[Article\]](#)
- [17] Po-Chedley, S., J.T. Fasullo, N. Siler, **Z.M. Labe**, E.A. Barnes, C.J.W. Bonfils, and B.D. Santer (2022). Internal variability and forcing influence model-satellite differences in the rate of tropical tropospheric warming. *Proceedings of the National Academy of Sciences*, DOI:10.1073/pnas.2209431119. [\[Article\]](#)
- [16] Witt, J.K., **Z.M. Labe**, and B.A. Clegg (2022). Comparisons of perceptions of risk for visualizations using animated risk trajectories versus cones of uncertainty. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, DOI:10.1177/1071181322661308. [\[Article\]](#)
- [15] Timmermans, M.-L. and **Z.M. Labe** (2022). [The Arctic] Sea surface temperature [in "State of the Climate in 2021"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-22-0082.1. [\[Article\]](#)
- [14] **Labe, Z.M.** and E.A. Barnes (2022). Comparison of climate model large ensembles with observations in the Arctic using simple neural networks. *Earth and Space Science*, DOI:10.1029/2022EA002348. [\[Article\]](#)
- [13] **Labe, Z.M.** and E.A. Barnes (2022). Predicting slowdowns in decadal climate warming trends with explainable neural networks. *Geophysical Research Letters*, DOI:10.1029/2022GL098173. [\[Article\]](#) [\[DOE Research Highlight\]](#)
- [12] Timmermans, M.-L. and **Z.M. Labe** (2021). Sea surface temperature [in "Arctic Report Card 2021"], NOAA, DOI:10.25923/2y8r-Oe49. [\[Article\]](#)

- [11] Timmermans, M.-L. and **Z.M. Labe** (2021), [The Arctic] Sea surface temperature [in "State of the Climate in 2020"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-21-0086.1. [\[Article\]](#)
- [10] **Labe, Z.M.** and E.A. Barnes (2021), Detecting climate signals using explainable AI with single-forcing large ensembles. *Journal of Advances in Modeling Earth Systems*, DOI:10.1029/2021MS002464. [\[Article\]](#)
- [9] Peings, Y., **Z.M. Labe**, and G. Magnusdottir (2021), Are 100 ensemble members enough to capture the remote atmospheric response to +2°C Arctic sea ice loss? *Journal of Climate*, DOI:10.1175/JCLI-D-20-0613.1. [\[Article\]](#) [\[CLIVAR\]](#)
- [8] Timmermans, M.-L. and **Z.M. Labe** (2020). Sea surface temperature [in "Arctic Report Card 2020"], NOAA, DOI:10.25923/vOfs-m920. [\[Article\]](#)
- [7] Timmermans, M.-L., **Z.M. Labe**, and C. Ladd (2020), [The Arctic] Sea surface temperature [in "State of the Climate in 2019"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-20-0086.1. [\[Article\]](#)
- [6] **Labe, Z.M.**, Y. Peings, and G. Magnusdottir (2020), Warm Arctic, cold Siberia pattern: role of full Arctic amplification versus sea ice loss alone, *Geophysical Research Letters*, DOI:10.1029/2020GL088583. [\[Article\]](#)
- [5] Thoman, R.L., U. Bhatt, P. Bieniek, B. Brettschneider, M. Brubaker, S. Danielson, **Z.M. Labe**, R. Lader, W. Meier, G. Sheffield, and J. Walsh (2020), The record low Bering Sea ice extent in 2018: Context, impacts and an assessment of the role of anthropogenic climate change [in "Explaining Extreme Events of 2018 from a Climate Perspective"]. *Bulletin of the American Meteorological Society*, DOI:10.1175/BAMS-D-19-0175.1 [\[Article\]](#)
- [4] **Labe, Z.M.**, Y. Peings, and G. Magnusdottir (2019), The effect of QBO phase on the atmospheric response to projected Arctic sea ice loss in early winter, *Geophysical Research Letters*, DOI:10.1029/2019GL083095. [\[Article\]](#)
- [3] **Labe, Z.M.**, Y. Peings, and G. Magnusdottir (2018), Contributions of ice thickness to the atmospheric response from projected Arctic sea ice loss, *Geophysical Research Letters*, DOI:10.1029/2018GL078158. [\[Article\]](#)
- [2] **Labe, Z.M.**, G. Magnusdottir, and H.S. Stern (2018), Variability of Arctic sea ice thickness using PIOMAS and the CESM Large Ensemble, *Journal of Climate*, DOI:10.1175/JCLI-D-17-0436.1. [\[Article\]](#)
- [1] **Labe, Z.M.**, T.R. Ault, and R. Zurita-Milla (2016), Identifying Anomalously Early Spring Onsets in the CESM Large Ensemble Project, *Climate Dynamics*, DOI:10.1007/s00382-016-3313-2. [\[Article\]](#)

Publications Submitted

- [4] Timmermans, M.-L. and **Z.M. Labe** (2026). [The Arctic] Sea surface temperature [in "State of the Climate in 2025"]. (*accepted*)
- [3] Zhang, Y. and **Z.M. Labe** (2026). Drivers of CO₂ Emissions and Climate Extremes. (*submitted*)
- [2] Witt, J.K., **Z.M. Labe**, and B.A. Clegg (2026). An Alternative to the "Cone of Uncertainty" that is Flexible, Intuitive, and Desirable for Communicating Hurricane Forecasts. (*submitted*)
- [1] **Labe Z.M.**, T.L. Delworth, N.C. Johnson, B.-T. Jong, C.E. McHugh, W.F. Cooke, and L. Jia (2025). Large reductions in United States heat extremes found in overshoot simulations with SPEAR, *EarthArXiv*, DOI:10.31223/X5TX4P (*submitted*) [\[Preprint\]](#)

Non-refereed/Other

- [8] Eayrs, C., L. Zampieri, L. von Albedyll, J. Brajard, M. Bushuk, S. Driscoll, I. Fenton, W. Gregory, L. Hoffman, W. De Jager, **Z.M. Labe**, X. Liu, F. Massonnet, A.

- McDonald, Y. Nie, M. Rogers, and T. Tian, April 2026: The ORCAS Community Workshop: Connecting Observations and AI Sea Ice Prediction Systems - Workshop Report. Zenodo, DOI: 10.5281/zenodo.19543215. [\[Article\]](#)
- [7] **Labe, Z.M.**, May 2023: Pan-Arctic Ice Ocean Modeling and Assimilation System (PIOMAS). Expert Contributor. NCAR Climate Data Guide. [\[Article\]](#)
- [6] **Labe, Z.M.**, August 2021: Sharing data-driven stories of Arctic climate change. WMO/WWRP Year of Polar Prediction, PolarPredictNews. [\[Article\]](#)
- [5] Peings, Y., **Z.M. Labe**, and G. Magnusdottir, August 2021: How reproducible is the response to +2°C Arctic sea-ice loss in a large ensemble of simulations? CLIVAR Research Highlight. [\[Article\]](#)
- [4] **Labe, Z.M.**, May 2020: The effects of Arctic sea-ice thickness loss and stratospheric variability on mid-latitude cold spells. University of California, Irvine. Doctoral dissertation. [\[PDF\]](#)
- [3] **Labe, Z.M.**, November 2019: Understanding Our Changing Arctic. Polar Bears International. Annual Magazine. [\[Article\]](#)
- [2] **Labe, Z.M.**, August 2017: Sea Ice Thickness Data Sets: Overview & Comparison Table. Expert Contributor. NCAR Climate Data Guide. [\[Article\]](#)
- [1] **Labe, Z.M.**, April 2015: Anomalously Early Onset of Spring in the CESM Large Ensemble. Cornell University. Undergraduate Honors Thesis. [\[PDF\]](#)

ORAL PRESENTATIONS

- [58] **Labe, Z.M.** Lecture II: Arctic Sea Ice, *CAMAS 2026 Early-Career School*, Monterey, CA (Apr 2026). **(Invited)**
- [57] **Labe, Z.M.** Applications of explainable machine learning for climate variability and change, *Department of Earth and Atmospheric Sciences*, Cornell University, Ithaca, NY (Mar 2026). **(Invited)**
- [56] **Labe, Z.M.** The urgency of effectively communicating climate data, *Climate Solutions Now 2026*, Remote Presentation (Feb 2026). **(Invited)**
- [55] **Labe, Z.M.** Visualizing our changing climate in real-time, *Terra.do*, Remote Presentation (Jul 2025). **(Invited-Keynote)**
- [54] **Labe, Z.M.** Monitoring our changing climate through data-driven stories, *8th C3S General Assembly*, Valencia, Spain (Jun 2025). **(Invited-Remote)**
- [53] **Labe, Z.M.** Utility of explainable machine learning for scenarios of climate change, Session III: AI Explainability and Climate Science Innovation, *Broadening Access to Climate AI Innovation Workshop*, University of Virginia, VA, USA (May 2025). **(Invited-Remote)**
- [52] **Labe, Z.M.** Using machine learning for understanding changes in U.S. summer-time temperatures, *NOAA Booth, 105th AMS Annual Meeting*, New Orleans, LA, USA (Jan 2025).
- [51] **Labe, Z.M.** XAI for detecting scenarios of future climate change, Climate & Global Dynamics Machine Learning Group, NSF NCAR, Boulder, CO, USA (Nov 2024). **(Invited-Remote)**
- [50] **Labe, Z.M.** Timing of emergence of U.S. summer temperatures with explainable AI, Building AI-Ready workforce for heat resilience, 6th NOAA AI Workshop, USA (Sep 2024). **(Invited-Remote)**
- [49] **Labe, Z.M.** Machine learning for climate prediction at decadal and longer timescales, *Predictability, Predictions, and Applications Interface (PPAI) Panel*, US CLIVAR, Boulder, CO, USA (Aug 2024). **(Invited-Remote)**
- [48] **Labe, Z.M.** Reexamining future projections of Arctic climate linkages, *Atmospheric and Oceanic Sciences Student/Postdoc Seminar*, Princeton University, NJ, USA (May 2024).

- [47] **Labe, Z.M.**, T.L. Delworth, N.C. Johnson, and W.F. Cooke. Explainable AI for distinguishing future climate change scenarios, *EGU General Assembly 2024*, Vienna, Austria (Apr 2024). [\[Abstract\]](#).
- [46] **Labe, Z.M.** Explainable AI approach for evaluating climate models in the Arctic. *Modelers' Community of Practice Webinar*, IARPC Collaborations (Mar 2024). **(Invited-Remote)**
- [45] **Labe, Z.M.** Explainable neural networks for evaluating patterns of climate change and variability. *Sharing Science - North American Webinar*, Young Earth System Scientists (YESS) Community (Mar 2024). **(Remote)**
- [44] **Labe, Z.M.** Applications of machine learning for climate change and variability. *Department of Environmental Sciences Seminar*, Rutgers University, New Brunswick, NJ (Feb 2024). **(Invited)**
- [43] **Labe, Z.M.**, T.L. Delworth, N.C. Johnson, and W.F. Cooke. A data-driven approach to identifying key regions of change associated with future climate scenarios, *23rd Conference on Artificial Intelligence for Environmental Science*, Baltimore, MD (Jan 2024). [\[Abstract\]](#)
- [42] **Labe, Z.M.**, N.C. Johnson, and T.L. Delworth. Distinguishing the regional emergence of United States summer temperatures between observations and climate model large ensembles, *23rd Conference on Artificial Intelligence for Environmental Science*, Baltimore, MD (Jan 2024). [\[Abstract\]](#)
- [41] **Labe, Z.M.** Revisiting projections of Arctic climate change linkages. *College of Surveying and Geo-informatics Seminar*, Tongji University, Shanghai, China (Nov 2023). **(Invited-Remote)**
- [40] **Labe, Z.M.** Using explainable machine learning to evaluate climate change projections. *Atmosphere and Ocean Climate Dynamics Seminar*, Yale University, CT (Oct 2023). **(Invited-Remote)**
- [39] **Labe, Z.M.** Creative machine learning approaches for climate change detection, *Resnick Young Investigators Symposium*, Caltech, NJ, USA (Apr 2023). **(Invited)**
- [38] **Labe, Z.M.** Using explainable AI to identify key regions of climate change in GFDL SPEAR large ensembles, *Lunchtime Seminar*, GFDL, Princeton, NJ, USA (Mar 2023).
- [37] **Labe, Z.M.** Using explainable machine learning for evaluating patterns of climate change, *Natural Sciences Group Seminar*, Washington State University Vancouver, WA, USA (Feb 2023). **(Invited-Remote)**
- [36] **Labe, Z.M.** Evaluating and communicating Arctic climate change projections, *Climate Change and Agriculture Guest Seminar*, Kansas State University, KS, USA (Feb 2023). **(Invited-Remote)**
- [35] **Labe, Z.M.** Exploring explainable machine learning for detecting changes in climate, *Department of Earth, Ocean, and Atmospheric Science Colloquium*, Florida State University, FL, USA (Feb 2023). **(Invited-Remote)**
- [34] **Labe, Z.M.** Forced climate signals with explainable AI and large ensembles, *Atmospheric and Oceanic Sciences Student/Postdoc Seminar*, Princeton University, NJ, USA (Feb 2023).
- [33] **Labe, Z.M.** Explainable AI for identifying regional climate change patterns, *Scientific Machine Learning Community*, University of Leeds, UK (Jan 2023). **(Invited-Remote)**
- [32] **Labe, Z.M.** and E.A. Barnes. Using artificial neural networks to predict temporary slowdowns in global warming trends, *22nd Conference on Artificial Intelligence for Environmental Science*, Denver, CO (Jan 2023). (Remote) [\[Abstract\]](#)

- [31] **Labe, Z.M.**, N.C. Johnson, and T.L. Delworth. Climate drivers of the recent spring-time cooling pattern in northern North America, *36th Conference on Climate Variability and Change*, Denver, CO (Jan 2023). (Remote) [\[Abstract\]](#)
- [30] **Labe, Z.M.** Machine learning for evaluating climate model projections, *Tech-Talks 2.0, IEEE-Student Branch - IIT Indore*, India (Dec 2022). (**Invited-Remote**)
- [29] **Labe, Z.M.** and E.A. Barnes. Using neural networks to predict temporary slow-downs in decadal climate warming trends, *27th Annual CESM Workshop*, Boulder, CO (Jun 2022). (Remote)
- [28] **Labe, Z.M.** Communicating Arctic climate change through data-driven stories, *Guest lecture for "Applications in Climate & Society"*, Columbia University, New York City, NY (Feb 2022). (**Invited-Remote**)
- [27] **Labe, Z.M.** Arctic climate change through the lens of data visualization, *Guest lecture for "Adaptation to Climate Change"*, North Carolina State University, Raleigh, NC (Jan 2022). (**Invited-Remote**)
- [26] **Labe, Z.M.** and E.A. Barnes. Using explainable neural networks for comparing climate model projections, *27th Conference on Probability and Statistics*, Virtual Attendance (Jan 2022). (Remote) [\[Abstract\]](#)
- [25] **Labe, Z.M.** and E.A. Barnes. Using neural networks to explore regional climate patterns in single-forcing large ensembles, *2021 AGU Annual Meeting*, New Orleans, LA (Dec 2021). (**Invited-Remote**) [\[Abstract\]](#)
- [24] **Labe, Z.M.** and E.A. Barnes. Evaluating global climate models using simple, explainable neural networks, *2021 AGU Annual Meeting*, New Orleans, LA (Dec 2021) (**Invited-Remote**) [\[Abstract\]](#)
- [23] **Labe, Z.M.** Exploring climate change signals with explainable AI. *Carbon Club*, NASA JPL, Pasadena, CA (Dec 2021). (**Invited-Remote**)
- [22] **Labe, Z.M.** and E.A. Barnes. Decadal warming slowdown predictions by an artificial neural network, *2021 Young Scientist Symposium on Atmospheric Research (YSSAR)*, Colorado State University, CO (Oct 2021).
- [21] **Labe, Z.M.** Assessing climate variability and change with explainable neural networks. *Lunchtime Seminar*, GFDL, Princeton University, NJ (Oct 2021). (**Invited-Remote**)
- [20] **Labe, Z.M.** and E.A. Barnes. Exploring climate model large ensembles with explainable neural networks, *WCRP workshop on attribution of multi-annual to decadal changes in the climate system*, Virtual Workshop (Sep 2021). (Remote)
- [19] **Labe, Z.M.** Learning new climate science by opening the machine learning black box. *Cognitive Brownbag Series*, Department of Psychology, Colorado State University, CO (Sep 2021). (**Invited**)
- [18] **Labe, Z.M.** and E.A. Barnes. Climate signals in CESM1 single-forcing large ensembles revealed by explainable neural networks, *26th Annual CESM Workshop*, Boulder, CO (Jun 2021). (Remote)
- [17] **Labe, Z.M.** Revealing climate change signals with explainable AI. *Spring Post-doctoral Research Symposium*, Colorado State University, CO (Mar 2021). (Remote)
- [16] **Labe, Z.M.** Communicating Arctic climate change through data-driven stories. *Arctic Science Summit Week 2021*, Lisbon, Portugal (Mar 2021). (**Invited-Remote**)
- [15] **Labe, Z.M.** The pan-Arctic impacts of thinning sea ice. Local Environmental Observer (LEO) Network, Anchorage, AK (Jan 2021). (**Invited-Remote**)
- [14] **Labe, Z.M.** and E.A. Barnes. Disentangling Climate Forcing in Multi-Model Large Ensembles Using Neural Networks. *20th Conference on Artificial Intelligence for Environmental Science*, Virtual Conference (Jan 2021). [\[Abstract\]](#)

- [13] **Labe, Z.M.** Why is it difficult to resolve future projections of Arctic-midlatitude linkages? *Atmospheric Dynamics Seminar*, Colorado State University, CO (Dec 2020). (Remote)
- [12] **Labe, Z.M.** Trends and regional variability of observed Arctic sea ice thickness. *ACCAP's Virtual Alaska Weather Symposia webinar series*, University of Alaska, Fairbanks, AK (Oct 2020). (**Invited-Remote**)
- [11] **Labe, Z.M.** Refining projections of the 'warm Arctic, cold Siberia' pattern in climate model simulations. *Atmosphere and Ocean Climate Dynamics Seminar*, Yale University, CT (Sep 2020). (**Invited-Remote**)
- [10] **Labe, Z.M.** Observations and climate model projections of Arctic climate change. Geography Department, Irvine Valley College, CA (May 2020). (**Invited-Remote**)
- [9] **Labe, Z.M.**, Y. Peings, and G. Magnusdottir. Detection of Signal in the Large-Scale Circulation Response to Arctic Sea-Ice Decline. *33rd Conference on Climate Variability and Change*, Boston, MA (Jan 2020). [[Abstract](#)]
- [8] **Labe, Z.M.** Melting Ice: Context, Causes, and Consequences of Polar Amplification. *Kavli Frontiers of Science, National Academy of Sciences*, Jerusalem, Israel (Sep 2019). (**Invited**)
- [7] **Labe, Z.M.** Projections of a future Arctic climate. Geography Department, Irvine Valley College, CA (May 2019). (**Invited**)
- [6] **Labe, Z.M.**, G. Magnusdottir, and Y. Peings. Linking the Quasi-Biennial Oscillation and projected Arctic sea-ice loss to stratospheric variability in early winter. *20th Conference on Middle Atmosphere*, Phoenix, AZ (Jan 2019). [[Abstract](#)]
- [5] **Labe, Z.M.**, Y. Peings, H.S. Stern, and G. Magnusdottir. Arctic sea ice thickness variability and its influence on the atmospheric response to projected sea ice loss. *Machine Learning and Physical Sciences (MAPS) Symposium*, University of California, Irvine (May 2018).
- [4] **Labe, Z.M.** Disentangling Arctic climate change and variability. Geography Department, Irvine Valley College, CA (Apr 2018). (**Invited**)
- [3] **Labe, Z.M.**, G. Magnusdottir, and H.S. Stern. Variability and future projections of Arctic sea ice thickness. *Understanding the Causes and Consequences of Polar Amplification Workshop*, Aspen Global Change Institute, Aspen, CO (Jun 2017).
- [2] **Labe, Z.M.** Communicating the future of Arctic climate change. Natural Sciences Division, Fullerton College, CA (Nov 2016). (**Invited**)
- [1] **Labe, Z.M.** Anomalously early onset of spring in the CESM Large Ensemble Project. *Earth and Atmospheric Science Undergraduate Research Symposium*, Cornell University, NY (May 2015).

POSTER PRESENTATIONS

- [12] **Labe, Z.M.**, and Y. Zhang. Designing Climate-Resilient Infrastructure with Extreme Event Attribution and Decadal Climate Prediction Systems, *21st Symposium on Societal Applications: Policy, Research and Practice*, Houston, TX (Jan 2026). [[Abstract](#)]
- [11] **Labe, Z.M.**, T.L. Delworth, N.C. Johnson, L. Jia, W.F. Cooke, B.-T. Jong, and C.E. McHugh. Greater reduction in U.S. heat extreme days in overshoot simulations with GFDL SPEAR, *38th Conference on Climate Variability and Change*, New Orleans, LA (Jan 2025). [[Abstract](#)]
- [10] **Labe, Z.M.**, L. Jia, N.C. Johnson, C.E. McHugh, T.L. Delworth, and W.F. Cooke. Prediction, projection, and detection of U.S. heat extremes using data-driven approaches with the GFDL SPEAR modeling system, *Extreme Heat Workshop: Emerging Risks from Concurrent, Compounding and Record-breaking Extreme Heat across Sectors*, Columbia University, New York City, NY (Jul 2024). [[Poster](#)]

- [9] **Labe, Z.M.**, N.C. Johnson, and T.L. Delworth. Identifying the drivers of the observed springtime cooling trend in northern North America with large ensemble simulations, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#) [\[Poster\]](#)
- [8] **Labe, Z.M.**, E.A. Barnes, and J. Hurrell. Detecting the regional emergence of climate signals with machine learning in a set of stratospheric aerosol injection simulations, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#) [\[Poster\]](#)
- [7] **Labe, Z.M.** and E.A. Barnes. Temporary slowdowns in decadal warming predictions by a neural network, *CLIVAR Climate Dynamics Panel (CDP) annual workshop: External versus internal variability on decadal and longer time scales*, Virtual Workshop (Oct 2022). [\[Poster\]](#)
- [6] **Labe, Z.M.** and E.A. Barnes. Climate model evaluation with explainable neural networks, *3rd NOAA Workshop on Leveraging AI in Environmental Sciences*, Virtual Workshop (Sep 2021). [\[Poster\]](#)
- [5] **Labe, Z.M.** and E.A. Barnes. Using explainable neural networks for comparing historical climate model simulations, *2nd Workshop on Knowledge Guided Machine Learning (KGML2021)*, Virtual Workshop (Aug 2021). [\[Poster\]](#)
- [4] **Labe, Z.M.** Loss of Arctic sea ice thickness affects the large-scale atmosphere, *Arctic System Change Workshop* at NCAR, Boulder, CO (Apr 2018). [\[Poster\]](#)
- [3] **Labe, Z.M.**, G. Magnusdottir, and H.S. Stern. Arctic Sea Ice Thickness Variability and the Large-scale Atmospheric Circulation Using Satellite Observations, PIOMAS, and the CESM Large Ensemble, *14th Conference on Polar Meteorology and Oceanography*, Seattle, WA (Jan 2017). [\[Abstract\]](#) [\[Poster\]](#)
- [2] **Labe, Z.M.**, G. Magnusdottir, and H.S. Stern. Making the most of Arctic sea ice thickness observations, *Symposium on Recent Advances in Data Science*, University of California, Irvine (Oct 2016). [\[Poster\]](#)
- [1] **Labe, Z.M.** and T.R. Ault. Anomalously Early Onset of Spring in the CESM Large Ensemble Project, *14th Annual AMS Student Conference*, Phoenix, AZ (Jan 2015). [\[Poster\]](#)

OTHER PRESENTATIONS

- [46] Jong, B.-T., **Z.M. Labe**, T.L. Delworth, and W.F. Cooke. Reversal of extreme precipitation trends over the Northeast US in response to aggressive climate mitigation, *EGU General Assembly*, Vienna, Austria (Apr 2025). (May 2026). [\[Abstract\]](#)
- [45] Joh, Y., S-W. Yeh, T.L. Delworth, **Z.M. Labe**, A.T. Wittenberg, W.F. Cooke, J. Lou, and Y-G. Park. Evolving synchronization of the Gulf Stream and Kuroshio-Oyashio Extension in a changing climate, *Ocean Sciences Meeting 2026*, Glasgow, Scotland (Feb 2026). [\[Abstract\]](#)
- [44] Mecray, E., **Z.M. Labe**, A. Smith, and J. Brady. Climate Central Science and Services to Support Information Needs Across Sectors, Hazards and Geographies, *30th Conference on Applied Climatology*, Houston, TX (Jan 2026). [\[Abstract\]](#)
- [43] Giguere, J., **Z.M. Labe**, D.M. Gilford, and K. Dahl. A Method for Rapid, Global Precipitation Attribution, *39th Conference on Climate Variability and Change*, Houston, TX (Jan 2026). [\[Abstract\]](#)
- [42] Ishraque, F., M.A. Shadab, O. Lauter, N. Narayanan, K. Shionalyn, M.N. Schaberg, **Z.M. Labe**, S.S. Gaikwad, M. Nicewonger, and R. Narayanan. Assessing challenges for polar early career scientists during science policy upheaval, *2025 AGU Annual Meeting*, New Orleans, LA (Dec 2025). [\[Abstract\]](#)
- [41] Kalashnikov, D.A., J. Abatzoglou, F.V. Davenport, **Z.M. Labe**, P.C. Loikith, D. Touma, and D. Signh. Projections of Lightning-Ignited Wildfire Risk in the Western United

States, *15th Fire and Forest Meteorology Symposium*, Denver, CO (May 2025). [\[Abstract\]](#)

- [40] Jong, B.-T., T.L. Delworth, **Z.M. Labe**, W.F. Cooke, and H. Murakami. Attributing changes in extreme precipitation across the Northeast U.S. under different climate scenarios, *EGU General Assembly*, Vienna, Austria (Apr 2025). [\[Abstract\]](#)
- [39] Schreck, C.J., **Z.M. Labe**, D.R. Easterling, R. Vose, J.E. Uehling, X. Zhang, D.A. Coates, N.C. Johnson, and J.J. Barsugli. Updates on NOAA's rapid attribution capability: The July 2024 California heat wave, *38th Conference on Climate Variability and Change*, New Orleans, LA (Jan 2025). [\[Abstract\]](#)
- [38] Kalashnikov, D.A., J. Abatzoglou, F.V. Davenport, **Z.M. Labe**, P.C. Loikith, D. Touma, and D. Signh. Projecting cloud-to-ground lightning and wildfire ignition risk in the western United States, *12th Conference on the Meteorological Application of Lightning Data*, New Orleans, LA (Jan 2025). [\[Abstract\]](#)
- [37] Jong, B.-T., T.L. Delworth, **Z.M. Labe**, and W.F. Cooke. Reversal of extreme precipitation trend over the Northeast U.S. in response to rapid reductions in greenhouse gas concentrations, *38th Conference on Climate Variability and Change*, New Orleans, LA (Jan 2025). [\[Abstract\]](#)
- [36] Kalashnikov, D.A., J.T. Abatzoglou, F.V. Davenport, **Z.M. Labe**, P.C. Loikith, D.E. Touma, and D. Singh. Projecting cloud-to-ground lightning and wildfire ignition risk in the western United States, *2024 AGU Annual Meeting*, Washington, DC (Dec 2024). [\[Abstract\]](#)
- [35] Bushuk, M., et al. (including **Z.M. Labe**). Predicting September Arctic Sea Ice: A Multi-Model Seasonal Skill Comparison, *2024 AGU Annual Meeting*, Washington, DC (Dec 2024). [\[Abstract\]](#)
- [34] Schaberg, M.N., **Z.M. Labe**, M.A. Shadab, L. Ashokkumar, S. Vanek, O. Lauter, and K. Shionalyn. Empowering the Polar Science Community: A Decade of USAPECS in Supporting Early Career Researchers Globally, *2024 AGU Annual Meeting*, Washington, DC (Dec 2024). [\[Abstract\]](#)
- [33] Johnson, N.C., M. Park, J. Clark, T.L. Delworth, L. Jia, F. Lu, W. Cooke, C. McHugh, and **Z.M. Labe**. Progress in the seasonal prediction of extreme weather in GFDL's SPEAR, *2024 AGU Annual Meeting*, Washington, DC (Dec 2024). [\[Abstract\]](#)
- [32] Park, M., N.C. Johnson, J. Clark, D. Yoon, **Z.M. Labe**, C. McHugh, and L. Jia. A Linkage between atmospheric rivers, blocking and North American heatwaves in high resolution climate model simulations, *2024 AGU Annual Meeting*, Washington, DC (Dec 2024). [\[Abstract\]](#)
- [31] Johnson, N.C., L. Jia, **Z.M. Labe**, C.E. McHugh, F. Lu, X. Yang, and T.L. Delworth. Diagnosing the predictability and simulation errors of seasonal extreme heat in the GFDL Seamless System for Prediction and Earth System Research (SPEAR), *NOAA's Subseasonal and Seasonal Applications Workshop*, College Park, MD, USA (Sep 2024). [\[Abstract\]](#)
- [30] Vanek, S., E. Adasheva, L. Ashokkumar, M.N. Helmberger, **Z.M. Labe**, O. Lauter, and M.A. Shadab. Exploring the past, present, and future of USAPECS: Lessons from a decade of supporting early career research across national and international polar networks, *Arctic Congress 2024*, Bodø, Norway (Jun 2024). [\[Abstract\]](#)
- [29] Kalashnikov, D.A., F.V. Davenport, **Z.M. Labe**, P.C. Loikith, J. Abatzoglou, and D. Singh. Using deep learning to predict cloud-to-ground lightning in the western United States, *23rd Conference on Artificial Intelligence for Environmental Science*, Baltimore, MD (Jan 2024). [\[Abstract\]](#)
- [28] Schreck, C.J., J. Barsugli, D.A. Coates, D.R. Easterling, K.E. Kunkel, **Z.M. Labe**, J.E. Uehling, R. Vose, and X. Zhang. Comparing the causes and unusualness of the Texas heatwaves in 2022 and 2023, *37th Conference on Climate Variability and Change*, Baltimore, MD (Jan 2024). [\[Abstract\]](#)

- [27] Ashokkumar, L, **Z.M. Labe**, M.A. Shadab, O. Lauter and E.P. Schreiber. Advancing inclusion, diversity, equity, and accessibility (IDEA) in the polar sciences by USAPECS, *2023 AGU Annual Meeting*, San Francisco, CA (Dec 2023). [\[Abstract\]](#)
- [26] Meem, T.J., **Z.M. Labe**, W.F. Cooke, T.L. Delworth, and V. Ramaswamy. Role of anthropogenic aerosols on the South Asian summer monsoon in a high-resolution large ensemble, *2023 AGU Annual Meeting*, San Francisco, CA (Dec 2023). [\[Abstract\]](#)
- [25] Kalashnikov, D.A., F.V. Davenport, **Z.M. Labe**, P.C. Loikith, J. Abatzoglou, and D. Singh. Using deep learning to predict cloud-to-ground lightning in the western United States, *2023 AGU Annual Meeting*, San Francisco, CA (Dec 2023). [\[Abstract\]](#)
- [24] Johnson, N.C., L. Jia, **Z.M. Labe**, T.L. Delworth, F. Lu, and C.E. McHugh. Sources of seasonal extreme heat predictability diagnosed from the GFDL seamless System for Prediction and Earth System Research (SPEAR), *2023 AGU Annual Meeting*, San Francisco, CA (Dec 2023). [\[Abstract\]](#)
- [23] J.J. Barsugli, **Z.M. Labe**, N.C. Johnson, K.E. Kunkel, T. Zhang, D.E. Easterling, J.E. Uehling, J. Eischeid, M. Hoerling, A. Kumar, D. Coates, R. Vose, D. Arndt, C.J. Schreck, and T.L. Delworth. The Texas heat wave of 2022: La Niña, drought, and the emerging climate change signal, *36th Conference on Climate Variability and Change*, Denver, CO (Jan 2023). [\[Abstract\]](#)
- [22] Johnson, N.C., T.L. Delworth, **Z.M. Labe**, F. Lu, and C.E. McHugh. Accurate seasonal predictions of the 2022 Texas early summer extreme heat nine months in advance, *36th Conference on Climate Variability and Change*, Denver, CO (Jan 2023). [\[Abstract\]](#)
- [21] Ashokkumar, L., L. Weinberg, **Z.M. Labe**, E. Schreiber, A. Taitt, and M. Dryak. Progress and challenges by early career polar scientists (USAPECS) in addressing inclusivity, diversity, equity, and accessibility, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#)
- [20] Po-Chedley, S., E.A. Barnes, C. Bonfils, J. Fasullo, **Z.M. Labe**, B. Santer, and N. Siler. Substantial contribution of internal variability to satellite-era tropospheric warming inferred from CMIP6 large ensembles, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#)
- [19] Lehner, F., C. Deser, J. Fasullo, E. Fischer, P. Hitchcock, **Z.M. Labe**, S. Milinski, M. Röthlisberger, I. Simpson, S. Sippel, and J. Zscheischler. Emergence of multiple feedbacks in the extreme and persistent warmth over Siberia in 2020, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#)
- [18] Bushuk, M., et al. (including **Z.M. Labe**). A multi-model comparison of September Arctic sea ice seasonal prediction skill, *2022 AGU Annual Meeting*, Chicago, IL (Dec 2022). [\[Abstract\]](#)
- [17] Witt, J.K., **Z.M. Labe**, and B. Clegg. Comparisons of Perceptions of Risk for Visualizations Using Animated Risk Trajectories Versus Cones of Uncertainty, *Human Factors and Ergonomics Society (HFES) 66th International Annual Meeting* (Oct 2022).
- [16] Po-Chedley, S., E.A. Barnes, C. Bonfils, J. Fasullo, **Z.M. Labe**, B. Santer, and N. Siler. Internal Variability and Forcing Influence Model-satellite Differences in the Rate of Tropical Tropospheric Warming, *Asia Oceania Geosciences Society 19th Annual Meeting* (Aug 2022).
- [15] Myint, H. and **Z.M. Labe**. Predicting September Arctic sea-ice using a hierarchy of statistical models, *21st Annual Student Conference: Polar Meteorology*, Virtual Attendance (Jan 2022). [\[Abstract\]](#)
- [14] Witt, J.K., **Z.M. Labe**, B. Clegg, and A. Warden. An alternative to the "Cone of Uncertainty" that is flexible, intuitive, and desirable for communicating hurricane

forecasts, *17th Symposium on Societal Applications: Policy, Research and Practice*, Virtual Attendance (Jan 2022). [\[Abstract\]](#)

- [13] Lehner, F., E. Fischer, **Z.M. Labe**, S. Milinski, M. Röthlisberger, I. Simpson, S. Sippel, and J. Zscheischler. Evaluating large ensembles for persistent extreme events such as the 2020 temperature anomaly over Siberia, *2021 AGU Annual Meeting*, Virtual Attendance (Dec 2021). [\[Abstract\]](#)
- [12] Peings, Y., **Z.M. Labe**, and G. Magnusdottir. Arctic-midlatitude linkages: role of sea ice loss versus full Arctic amplification. *US CLIVAR PPAI Panel Webinar Series*, Virtual Talk (Apr 2021).
- [11] Peings, Y., **Z.M. Labe**, and G. Magnusdottir. Influence of internal variability: how to ensure results are robust? *Polar Amplification Model Intercomparison (PAMIP) Virtual Workshop*, Virtual Conference (Mar 2021).
- [10] Peings, Y., **Z.M. Labe**, and G. Magnusdottir. Arctic-midlatitude linkages: role of sea ice loss versus full Arctic Amplification. *Arctic Science Summit Week 2021*, Virtual Conference (Mar 2021).
- [9] Magnusdottir, G., **Z.M. Labe**, and Y. Peings. The midlatitude response to Arctic sea-ice decline, compared to the response to the full effects of Arctic amplification. *34th Conference on Climate Variability and Change*, Virtual Conference (Jan 2021). [\[Abstract\]](#)
- [8] Peings, Y., G. Magnusdottir, and **Z.M. Labe**. Are 100 ensemble members enough to capture the remote atmospheric response to +2°C Arctic sea ice loss? *34th Conference on Climate Variability and Change*, Virtual Conference (Jan 2021). [\[Abstract\]](#)
- [7] Magnusdottir, G., **Z.M. Labe**, and Y. Peings. How does the atmospheric response to Arctic sea-ice decline compare to the full effect of the Arctic Amplification? *2020 AGU Annual Meeting*, Virtual Conference (Dec 2020). [\[Abstract\]](#)
- [6] Peings, Y., **Z.M. Labe**, and G. Magnusdottir. Are 100 ensemble members enough to capture the remote atmospheric response to +2°C Arctic sea ice loss? *2020 AGU Annual Meeting*, Virtual Conference (Dec 2020). [\[Abstract\]](#)
- [5] Magnusdottir, G., Y. Peings, and **Z.M. Labe**. Response to sea-ice loss under the Polar Amplification MIP protocol in extended ensembles of simulations. *2019 AGU Annual Meeting*, San Francisco, CA (Dec 2019). [\[Abstract\]](#)
- [4] Peings, Y., G. Magnusdottir, and **Z.M. Labe**. Impact of the QBO on the response to Arctic sea ice loss. *Polar Amplification Model Intercomparison (PAMIP) Workshop*, Devon, UK (Jun 2019).
- [3] Holman, A., R. Thoman, **Z.M. Labe**, and J.E. Walsh. Not Just Chance: Ocean and Atmospheric Factors in the Record Low Bering Sea Ice Winter of 2017-2018 and effects on health and safety, *2018 AGU Annual Meeting*, Washington, DC (Dec 2018). [\[Abstract\]](#)
- [2] Magnusdottir, G., **Z.M. Labe**, and Y. Peings. The role of the stratosphere, including the QBO, in Arctic to mid-latitude teleconnections associated with sea-ice forcing. *2018 AGU Annual Meeting*, Washington, DC (Dec 2018). [\[Abstract\]](#)
- [1] Thoman, R. and **Z.M. Labe**. 2017-18 Sea Ice in Western Alaska during the 2017-18 Season: Historical Context and Possible Drivers. *Western Alaska Interdisciplinary Science Conference and Forum*, Nome, AK (Mar 2018).

AWARDS AND FELLOWSHIPS

Sep. 2025	Outstanding Reviewer Award - <i>JGR: Machine Learning</i>
Mar. 2025	Outstanding Reviewer Award - <i>Environmental Research: Climate</i>
Sep. 2023	Outstanding Reviewer Award - <i>Geophysical Research Letters</i>

Oct. 2022	Early Career Scientist Best Poster Award - CLIVAR CDP 2022 Workshop: External versus internal variability on decadal and longer time scales
May 2021	Outstanding Reviewer Award - <i>Geophysical Research Letters</i>
2020-2021	SoGES Sustainability Leadership Fellow, Colorado State University
Sep. 2019	Kavli Fellow, National Academy of Sciences
Jan. 2019	Outstanding Student Presentation Award - AMS Conference on Middle Atmosphere
Nov. 2018	Travel Award - AMS Conference on Middle Atmosphere
2016-2018	NSF NRT-DESE in Data Science and Physical Science Fellowship
Su. 2016	Data Science Initiative Fellowship, Department of Statistics, University of California, Irvine
Wi. 2016	Jenkins Family Graduate Fellowship in Earth System Science, Department of Earth System Science, University of California, Irvine
2014-2015	Fuerst Outstanding Library Student Employee Award Finalist, Cornell University

TEACHING AND OUTREACH

Mentoring Experience

Su. 2023	Tasmeem Jahan Meem - CIMES Research Internship Program at Princeton University
Su. 2021	Hannah Myint - REU at Colorado State Univ. (Dept. of Atmos.)

University of California, Irvine

Fa. 2019	Teaching Assistant, Terrestrial Hydrology
Sp. 2019	Teaching Assistant, Earth System Physics
Wi. 2019	Teaching Assistant, Weather Analysis
Fa. 2016	Teaching Assistant, Fundamental Processes in Earth and Environmental Studies

Cornell University

Su. 2015	Guest Lecturer, Severe Weather Phenomena
Sp. 2015	Teaching Assistant, Programming and Meteorology Software (Python)
Fa. 2014	Teaching Assistant, Basic Principles of Meteorology Lab

Related Educational Activities and Talks

May 2026	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from April 2026. (Presenter-Remote)
Apr. 2026	<i>USAPECS</i> , <i>Community Engagement Training Course</i> , Making accessible data visualization. (Invited-Presenter-Remote).
Apr. 2026	<i>ANTH/SOCI 3320: Indigenous Cultures of North America</i> , Introduction to rapid Arctic change. The University of Texas at El Paso, El Paso, TX. (Invited-Guest Lecture-Remote).

Apr. 2026	<i>Pennswood Village</i> , The rapidly changing Arctic and its far-reaching impacts, Newtown, PA. (Invited-Presenter)
Apr. 2026	<i>GEO 371T/391: Climate Data</i> , Creative and effective climate data visualization. The University of Texas at Austin, Austin, TX. (Invited-Guest Lecture-Remote) .
Apr. 2026	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from March 2026. (Presenter-Remote)
Mar. 2026	<i>American Camp Association, Climate Symposium</i> , Ask a meteorologist: Extreme weather and the climate change connection. (Invited-Organizer-Remote) .
Mar. 2026	<i>Covering Climate Now, Press Briefing</i> , "Virtually Impossible Heat" & the Future of the American West. (Invited-Panelist-Remote) .
Mar. 2026	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from February 2026. (Presenter-Remote)
Mar. 2026	<i>World Water Week 2026 at UTEP and UACJ</i> , Climate change and human adaptations in the Arctic, El Paso, TX. (Invited-Panelist-Remote) .
Mar. 2026	<i>Cornell University, Climate Change Seminar</i> , Connecting global climate change to local impacts, Ithaca, NY. (Invited-Presenter) .
Mar. 2026	<i>National Garden Clubs</i> , Environmental School Course 3, St. Louis, MO. (Invited-Presenter-Remote) .
Feb. 2026	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from January 2026. (Presenter-Remote)
Feb. 2026	<i>Horticultural Societies of Parkdale and Toronto</i> , Climate change extremes by the season, Toronto, Canada. (Invited-Presenter-Remote) .
Jan. 2026	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from December 2025. (Presenter-Remote)
Jan. 2026	<i>NOAA Global Ocean Monitoring and Observing</i> , Committee on Marine Transportation System Dialogue. (Invited-Presenter-Remote) .
Dec. 2025	<i>Climate Central</i> , Climate Central's Monthly Briefing Highlights from November 2025. (Presenter-Remote)
Nov. 2025	<i>Cooperative Atmospheric and Earth Science Assemble of Researchers</i> , Sharing complex research through effective communication techniques. (Invited-Presenter-Remote) .
Nov. 2025	<i>EV228: Analysis of Environmental Data</i> , Data visualization for trusted science communication, Colorado College, Colorado Springs, CO. (Invited-Guest Lecture-Remote) .
Aug. 2025	<i>Climate Central Lunch & Learn</i> , Bringing real-time climate monitoring to life. (Presenter-Remote) .
Jul. 2025	<i>COLDEX Research Experience for Undergraduates</i> , Earth science careers. (Panelist-Invited-Remote)
Jul. 2025	<i>The Medical Society Consortium on Climate and Health</i> , Harnessing the Power of Social Media: Using Different Platforms Strategically. (Panelist-Invited-Remote)
Jun. 2025	<i>The Weather and Climate Livestream</i> , Science in retreat: The consequences for the Arctic and our planet, Remote Presentation. (Invited Keynote)
Apr. 2025	<i>EGU General Assembly 2025</i> , Academic activism for urgent climate action: How to get started? Vienna, Austria. (Presenter-Invited)
Apr. 2025	<i>Pennswood Village</i> , Our changing climate in the Northeastern United States, Newtown, PA. (Invited-Presenter)

Zachary Labe

Feb. 2025	<i>Rider University, Global Biogeochemistry Class Visit, The Arctic through the lens of data visualization, NOAA GFDL, Princeton, NJ. (Invited)</i>
Feb. 2025	<i>Guest lecture for "First-Year Innovation and Research Experience and Climate Computing", Using machine learning for understanding changes in the Earth system, University of Maryland, MD (Invited-Remote)</i>
Feb. 2025	<i>18th Annual Monmouth Junction Elementary School PTO Science Fair, Meet a scientist event, Dayton, NJ (Volunteer)</i>
Jan. 2025	<i>Canadian Meteorological and Oceanographic Society, Virtual workshop for "How are technological and climate disruptions impacting education and careers in the weather and climate enterprise?" (Panelist-Invited-Remote)</i>
Jan. 2025	<i>NOAA Research, NOAA Constituent Briefing on the 2024 Arctic Report Card. (Panelist-Invited-Remote)</i>
Jan. 2025	<i>NOAA Library, Two decades of the NOAA Arctic Report Card: history, reflections, and next steps. (Panelist-Invited-Remote)</i>
Jan. 2025	<i>Joint with 38th AMS Conference on Climate Variability and Change and 34th AMS Conference on Education, Session convener for "Communicating climate change uncertainty to the public," New Orleans, LA.</i>
Oct. 2024	<i>NOAA Climate Connections 2.0, Session convener for "Back-casting Climate Projection Services: Enhancing NOAA's Climate Services Culture," Washington, DC.</i>
Aug. 2024	<i>EOS Subject Matter Expert, Earth's atmosphere illustration for AGU publications</i>
Jul. 2024	<i>GFDL/AOS Summer Internship Lecture Series, Data Visualization Workshop, Princeton University, Princeton, NJ. (Presenter)</i>
May 2024	<i>Arctic Congress 2024, Session chair for "Cross/Trans-Disciplinary Collaborations and Advancing Community, Equity, and Inclusion in Arctic Research."</i>
May 2024	<i>21st Annual New Jersey Ocean Fun Days, Volunteer for atmospheric and oceanic sciences outreach booth for CIMES, Island Beach State Park/Sandy Hook, NJ. (Presenter)</i>
Apr. 2024	<i>United States Association of Polar Early Career Scientists (US-APECS) IDEA Training Course, Accessibility and disability in on-line spaces. (Presenter-Remote)</i>
Mar. 2024	<i>GFDL Polar Climate Interest Group, Machine learning in the cryosphere, Princeton University, Princeton, NJ. (Presenter)</i>
Mar. 2024	<i>Temple University, Climate Communication Workshop: Learn How To Make Your Research Matter, Using accessible data to communicate global climate change, Philadelphia, PA. (Invited Keynote)</i>
Feb. 2024	<i>17th Annual Monmouth Junction Elementary School PTO Science Fair, Meet a scientist event, Dayton, NJ (Volunteer)</i>
Jan. 2024	<i>AMS AI Conference - Student Competition, Poster Judge</i>
Jan. 2024	<i>37th AMS Conference on Climate Variability and Change, Session chair for "Frontiers in Earth System Modeling," Baltimore, MD.</i>
Jan. 2024	<i>ACCAP - IARC, University of Alaska Fairbanks, Arctic Report Card: Background and Key Finding. (Panelist-Invited-Remote)</i>
Dec. 2023	<i>NOAA Research, 2023 Arctic Report Card Stakeholder Briefing. (Panelist-Invited-Remote)</i>

Dec. 2023	<i>Mercer County Community College</i> , Career pathways and re- search opportunities in the Earth sciences, West Windsor, NJ.
Nov. 2023	<i>NJ State Museum Planetarium</i> , Visualizing climate change through data, Trenton, NJ. (Invited)
Oct. 2023	<i>Euronews</i> , Climate Now debate: 2023 is set to be the hottest year on record, so why aren't we taking action? (Panelist- Invited-Remote)
Sep. 2023	Guest lecture for "Observing and Modeling Climate Change (EES 3506/5506)", Contrasting polar climate change in the past, present, and future, Temple University, PA. (Invited-Remote)
Sep. 2023	<i>Hershey Horticulture Society</i> , Climate change extremes by sea- son in the United States, Hershey, PA. (Presenter-Invited) .
Aug. 2023	Guest lecture for "Introduction to Global Climate Change (ESS 15)", Our changing Arctic in the past and future, University of Califor- nia, Irvine, CA. (Invited-Remote)
Jun. 2023	<i>La Uni Climática</i> , Monitoring indicators of climate change through data-driven visualization. <i>(Presenter-Invited-Remote)</i>
Jun. 2023	<i>EUMET/Copernicus Earth Observation - Data Visualization Work- shop Series</i> , Climate Extremes: Heatwaves, Changes in Ice, Drought, Sea Ice Anomalies. <i>(Presenter-Invited-Remote)</i>
Apr. 2023	<i>Mercer County Community College</i> , Career pathways and re- search opportunities in the Earth sciences, West Windsor, NJ.
Feb. 2023	<i>Rider University</i> , <i>Global Biogeochemistry Class Visit</i> , Arctic cli- mate change through the lens of data visualization, NOAA GFDL, Princeton, NJ. (Invited) .
Dec. 2022	<i>AGU Fall Meeting 2022</i> , Outstanding student presentation awards judge (OSPA)
Nov. 2022	<i>GFDL Lunchtime Seminar Series</i> , Using data visualization for ac- cessible science (communication), Princeton, NJ. (Presenter- Remote) .
Oct. 2022	<i>IEEEVis 2022</i> , #Viz4Climate - Workshop on High-Impact Tech- niques for Visual Climate Science Communication, Oklahoma City, OK. (Panelist-Invited-Remote)
Jul. 2022	<i>REU Professional Development Series</i> , Making effective and ac- cessible figures, Colorado State University, Fort Collins, CO. (Presenter-Invited-Remote)
Jul. 2022	<i>Polar Bears International</i> , 'State of the Arctic 2022' - Blog Post (Author) [Article]
Jun. 2022	<i>GFDL/AOS Summer Internship Lecture Series</i> , Learning new cli- mate science by thinking creatively with machine learning, Princeton University, Princeton, NJ. (Presenter)
Apr. 2022	Guest lecture for "Objective Analysis in Atmospheric Sciences (ATS 655)", Using data visualization for effective science (communi- cation), Colorado State University, Fort Collins, CO. (Remote)
Nov. 2021	<i>CASSINI Hackathon</i> , Climate Change in the Arctic, Nordic Startup School, Finland. (Presenter-Invited)
Jul. 2021	<i>REU Professional Development Series</i> , Making effective figures, Colorado State University, Fort Collins, CO. (Presenter-Invited)
Jul. 2021	<i>Polar Bears International</i> , 'State of the Arctic' - Blog Post (Author) [Article]
Jun. 2021	<i>US Climate Variability and Predictability</i> , 2021 June CLIVAR Newsgram (Featured Graphics) [Article]
Jun. 2021	<i>ABC News</i> , Climate summit for the ABC network (Panelist- Invited-Remote)

Zachary Labe

May 2021	Guest lecture for "Objective Analysis in Atmospheric Sciences (ATS 655)", Using data visualization for effective science (communication), Colorado State University, Fort Collins, CO. (Remote)
Mar. 2021	Department of Atmospheric Science, Improving science (communication) through data visualization, Colorado State University, Fort Collins, CO. (Presenter-Remote)
Feb. 2021	School of Global Environmental Sustainability - HumanNature Blog, 'Telling stories with data' - Blog Post (Author) [Article]
Jan. 2021	20th AMS Annual Student Conference, Poster Judge (Remote)
Oct. 2020	Cornell Chapter of the American Meteorological Society, Professional Series: Academic Sector (Panelist-Invited-Remote)
Oct. 2020	AMS Early Career Leadership Academy, Session on Climate Change Communications (Panelist-Invited-Remote)
Sep. 2020	Polar Bears International, 'A Sign of the Future: Summer 2020 in the Arctic' - Blog Post (Author) [Article]
Jul. 2020	Polar Bears International, 'State of the Arctic in 2020' - Blog Post (Author) [Article]
Mar. 2020	Climate, Literacy, Empowerment And iNquiry (CLEAN), The Cryosphere Outreach event, MacArthur Fundamental Intermediate School, Santa Ana, CA (Presenter)
Feb. 2020	Irvine Unified School District's 39 th Annual Science Fair, Physical science project judge, Irvine, CA (Volunteer)
Jan. 2020	100th American Meteorological Society Annual Meeting: Side Panel - Using Social Media to Communicate Climate Science, Boston, MA. (Chair) [Abstract]
Nov. 2019	Department of Earth System Science, Improving scientific graphics of climate (change) data, University of California, Irvine, CA. (Presenter)
Oct. 2019	Irvine Unified School District, "Ask-A-Scientist/Engineer Night" at Creekside High School, Orange County, CA (Volunteer)
Sep. 2019	Geological Society of America Annual Meeting, Geoscience Communication in the Modern Age – "A Changing Arctic", Phoenix, AZ (Contributed Art)
Sep. 2019	UC Irvine Graduate Orientation, Session Title – Healthy Student Life – The Current Student Perspective, Irvine, CA (Panelist-Invited)
May 2019	College and Career Day Experience, MacArthur Fundamental Intermediate School, Santa Ana, CA (Presenter)
Apr. 2019	Vista Verde School, Weather and climate booth at an Earth Science Day fair, Irvine, CA (Presenter)
Mar. 2019	Climate, Literacy, Empowerment And iNquiry (CLEAN), Weather and Climate Outreach event, MacArthur Fundamental Intermediate School, Santa Ana, CA (Presenter)
Feb. 2019	Orange County Regional Science Olympiad, Climate Science-Meteorology exam writer and proctor, Irvine, CA (Event Supervisor)
Feb. 2019	Climate, Literacy, Empowerment And iNquiry (CLEAN), The Cryosphere Outreach event, MacArthur Fundamental Intermediate School, Santa Ana, CA (Presenter)
Dec. 2018	NOAA, Arctic Report Card - Update for 2018. [Contributed Graphics]
Oct. 2018	Brews and Brains: TED-Style Talk, Our Changing Arctic, Fireside Tavern, Costa Mesa, CA. (Presenter-Invited).

Oct. 2018	<i>Irvine Unified School District, "Ask-A-Scientist/Engineer Night" at Rancho San Joaquin Middle School, Orange County, CA</i> (Volunteer)
Sep. 2018	<i>UC Irvine Graduate Orientation, Session Title – Student Panel – Embracing Your New Graduate Life, Irvine, CA</i> (Panelist-Invited)
May 2018	<i>Vista Verde School, Weather and climate booth at an Earth Day fair, Irvine, CA</i> (Presenter)
Feb. 2018	<i>Irvine Unified School District's 37th Annual Science Fair, Physical science project judge, Irvine, CA</i> (Volunteer)
Feb. 2018	<i>Climate, Literacy, Empowerment And iNquiry (CLEAN), Weather and Climate Outreach event, MacArthur Fundamental Intermediate School, Santa Ana, CA</i> (Presenter)
Feb. 2018	<i>Orange County Regional Science Olympiad, Climate Science-Meteorology exam writer and proctor, Irvine, CA</i> (Event Supervisor)
Feb. 2018	<i>Climate, Literacy, Empowerment And iNquiry (CLEAN), The Cryosphere Outreach event, MacArthur Fundamental Intermediate School, Santa Ana, CA</i> (Presenter)
Sep. 2017	<i>Irvine Unified School District, "Ask-A-Scientist/Engineer Night" at Rancho San Joaquin Middle School, Orange County, CA</i> (Volunteer)
Aug. 2017	<i>Irvine Unified School District, "Ask-A-Scientist/Engineer Night" at Rancho San Joaquin Middle School, Orange County, CA</i> (Volunteer)
Aug. 2017	<i>NASA DIRECT-STEM Program, Assessing a changing Arctic</i> (Presenter-Invited) .
Apr. 2017	<i>Climate Data Hackathon, Looking for clues to changes in Arctic sea ice, University of California, Irvine.</i> (Presenter-Invited) .
Feb. 2017	<i>Irvine Unified School District's 36th Annual Science Fair, Physical science project judge, Irvine, CA</i> (Volunteer)
Feb. 2017	<i>Orange County Regional Science Olympiad, Severe Storms-Meteorology exam writer and proctor, Irvine, CA</i> (Event Supervisor)
Wi. 2017	<i>Seminar Course – Teaching Topics in Earth System Science</i> (Training)
Oct. 2016	<i>Irvine Unified School District, "Ask-A-Scientist/Engineer Night" at Rancho San Joaquin Middle School, Orange County, CA</i> (Volunteer)
Aug. 2016	<i>NASA DIRECT-STEM Workshop, Data and Analysis of Arctic Climate Change in CMIP5 using Python</i> (Instructor) .

PROFESSIONAL ACTIVITIES & SERVICE

Summary –	Conference Paper Referee - <i>AI4Earth NeurIPS, ICML ML4ESM</i> Hiring Committees - <i>Climate Central Inc., NOAA OAR</i> Proposal Referee - <i>NOAA (CPO/MAPP), NSF (GEO/AGS), NSF (ANS), Netherlands Organisation for Scientific Research (NWO), S.C. Sea Grant Consortium</i>
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	Journal Manuscript Referee [Web of Science ; >100 reviews] - <i>Artificial Intelligence for the Earth Systems, Atmosphere-Ocean, Atmospheric Research, Atmosphere, Bulletin of the American Meteorological Society, Climatic Change, Climate Dynamics, Communications Earth & Environment, Earth's Future, Environmental Data Science, Environmental Research: Climate, Environmental Research Letters, Geophysical Research Letters, Geosciences, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, International Journal of Climate, International Journal of Disaster Risk Science, Journal of Climate, JGR: Atmospheres, JGR: Machine Learning and Computation, Journal of Advances in Modeling Earth Systems, Journal of Glaciology, Nature Climate Change, Nature Communications, Polar Geography, Remote Sensing, Science Advances, Scientific Data, Scientific Reports, The Cryosphere, Weather, Weather and Climate Dynamics</i>
2026-Present	Subject Matter Expert - AMS Certificate in Climate Science
2026-Present	Board Member - AMS Committee on Climate Services
2025-Present	Associate Editor - <i>Atmospheric Science Letters</i>
2025-Present	Task Team - WMO's WWRP, PCAPS/SCOR ORCAS
2023-Present	Associate Editor - <i>Journal of Climate</i>
2020-Present	Board of Advisors for the NCAR Climate Data Guide
2023-2025	Member - Fresh Eyes on CMIP (WCRP Working Group)
2021-2025	Board Member - US Association of Polar Early Career Scientists
2023-2025	Contributing Editor - Carbon Brief
2023-2025	NOAA GFDL Internal Reviewer
2024-2025	Board Member - NOAA GFDL Employees Association (GFDLEA)
2023-2025	Member - Diversity, Equity, Inclusivity, and Accessibility Committee (NOAA GFDL)
2021-2024	Guest Editor - Special Issue for <i>Atmospheric Science Letters</i>
Jan. 2024	Assessment of the Algorithm Reviewer - EUMETSAT/OSI SAF Sea Ice Index (OSI-420)
Nov. 2023	Review Panel Member - NASA
Apr. 2023	Review Panel Member - NOAA
Nov. 2022	Data Requirement Review Panel Member - EUMETSAT/OSI SAF Sea Ice Index
Oct. 2022	Review Panel Member - NSF
2021-2022	Science Associate/Volunteer - Arctic Basecamp
2019-2021	Student Board Member, American Meteorological Society's Board on Societal Impacts
2017-2020	Vice President & Communications Chair, Student Chapter of the American Meteorological Society at the University of California, Irvine
2017-2020	Student Board Member, American Meteorological Society's Committee on Climate Variability and Change
2017-2018	Member, Student Communications Working Group, University of California, Irvine
2014-2015	Co-President, American Meteorological Society Student Chapter, Cornell University

PROFESSIONAL TRAVEL

Affiliations/Memberships

2023-Present	European Geosciences Union, Member
2021-Present	Interagency Arctic Research Policy Committee (IARPC), Member
2020-Present	American Association for the Advancement of Science (AAAS), Member
2016-Present	Association of Polar Early Career Scientists (APECS), Member
2015-Present	American Geophysical Union (AGU), Member
2014-Present	National Weather Association (NWA), Member
2009-Present	American Meteorological Society (AMS), Member

Field Work

May 2017	INTPART Arctic Field Summer Schools: Norway–Canada–USA collaboration, University of Tromsø, Norway (Workshops); Longyearbyen, Svalbard (R/V <i>Lance</i> - sea ice)
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Workshops

Apr. 2026	CAMAS 2026 Workshop and Early-Career School, Monterey, CA
Feb. 2026	PCAPS/SCOR ORCAS, Community Workshop: Connecting Observations and AI Sea Ice Prediction Systems. (Virtual)
Jan. 2026	AMS Houston Area Heat Workshop, Houston, TX
Jun. 2025	Natural Hazard Risk Reduction: Building the Resiliency Business Case, ASCE-NOAA workshop. (Virtual)
Jun. 2025	AASC Climate Tools "Boot Camp," Blue Springs, MO. (Virtual)
May 2025	Broadening Access to Climate AI Innovation Workshop, University of Virginia, VA. (Virtual)
Feb. 2025	Arctic Report Card Planning Workshop, Seattle, WA. (Virtual)
Nov. 2024	Data-Driven Precision Health: Integrating Genomics, Climate, and the Environment Conference, Princeton, NJ
Oct. 2024	NOAA Climate Connections 2.0, Washington, DC. (Virtual)
Oct. 2024	PAMIP Phase 2 Workshop. (Virtual)
Jul. 2024	Extreme Heat Workshop: Emerging Risks from Concurrent, Compounding and Record-breaking Extreme Heat across Sectors, New York City, NY
Jun. 2024	Climate Risk Reduction ASCE-NOAA Workshop: Hazards and Process. (Virtual)
Aug. 2023	Princeton AOS's Paleo, present, and future: Leveraging the past to understand and predict our changing climate, Princeton, NJ
Oct. 2022	IEEE VIS 2022, Oklahoma City, OK. (Virtual)
Oct. 2022	CLIVAR Climate Dynamics Panel (CDP) annual workshop: External versus internal variability on decadal and longer time scales. (Virtual)
Sep. 2022	Leveraging Earth System Science and Modeling to Inform Civil Engineering Design: ASCE-NOAA. (Virtual)
Aug. 2022	Princeton AOS's Workshop on Attribution of Extreme Events to Climate Change, Princeton, NJ

Jul. 2022	The Reproducibility Crisis in ML-based Science, Princeton University. (Virtual)
Jun. 2022	2022 CESM Workshop, National Center for Atmospheric Research (NCAR), Boulder, CO. (Virtual)
Sep. 2021	WCRP Workshop on Attribution of Multi-Annual to Decadal Changes in the Climate System. (Virtual)
Sep. 2021	3rd NOAA Workshop on Leveraging AI in Environmental Sciences. (Virtual)
Aug. 2021	2nd Workshop on Knowledge Guided Machine Learning (KGML2021). (Virtual)
Jun. 2021	2021 CESM Workshop, National Center for Atmospheric Research (NCAR), Boulder, CO. (Virtual)
May 2021	NOAA Arctic Report Card Workshop 2021. (Virtual)
Jun. 2020	Artificial Intelligence for Earth System Science (AI4ESS) Summer School, UCAR/NCAR. (Virtual)
Oct. 2019	Transdisciplinary Research on the Changing Arctic and Its Global Impacts: Enhancing Capacity for Convergence Science, Beckman Center of the NAS, Irvine, CA
Apr. 2018	Arctic System Change Workshop, Mesa Lab, National Center for Atmospheric Research (NCAR), Boulder, CO
Jun. 2017	Understanding the Causes and Consequences of Polar Amplification, Aspen Global Change Institute, Aspen, CO

TECHNICAL SKILLS

Programming: Python, MATLAB, R, Fortran, Unix, HTML, $\text{\TeX}$, $\text{\LaTeX}$, $\text{\BibTeX}$
Software/Tools: NCL, NCO, CDO, GrADS, Git, Scikit-learn, TensorFlow/Keras
Modeling: NCAR's Yellowstone/Cheyenne, CESM, E3SM, CMIP5/6

Visit my [GitHub](#) (>300 followers) !

SELECTED NEWS AND MEDIA

May 2026	<i>Farm Progress</i> , "Frozen out: Pennsylvania grower starts long slog to orchard recovery" (Quoted). [Article]
May 2026	<i>LA Times</i> , "This coming El Niño could be a monster. Will it bring epic rain to California this winter?" (Quoted). [Article]
May 2026	<i>Scientific American</i> , "Carbon dioxide levels in the atmosphere just hit a 'depressing' record high" (Quoted). [Article]
Apr. 2026	<i>High Country News</i> , "The ramifications of record-shattering heat on the West's ecosystems" (Quoted). [Article]
Apr. 2026	<i>Dr Gilbz</i> , "The 'grim news' that isn't making headlines" (Interview). [Recording]
Apr. 2026	<i>Grist</i> , "Deep-diving robots help crack the mystery of Antarctica's vanishing sea ice" (Quoted). [Article]
Apr. 2026	<i>E&E News</i> , "NOAA halts crucial dataset that helps measure Arctic sea ice" (Quoted). [Article]
Mar. 2026	<i>Aspen Public Radio</i> , "March heat wave is 'particularly alarming,' but not our new normal, yet" (Interview). [Article]
Mar. 2026	<i>Carbon Brief</i> , "'Very alarming' winter sees Arctic sea ice hit record-low for second year running" (Quoted). [Article]

Mar. 2026	<i>Heatmap News</i> , "A Once-in-a-4,433-Year Heat Wave Is Hitting the Western U.S." (Quoted) . [Article]
Mar. 2026	<i>Nevada Public Radio</i> , "Spring is arriving earlier due to climate change" (Quoted) . [Article]
Feb. 2026	<i>Carbon Brief</i> , "Q&A: How Trump is threatening climate science in Earth's polar regions" (Quoted) . [Article]
Jan. 2026	<i>Delphi Zero</i> , "Letting the Ice Speak" (Interview) . [Article]
Jan. 2026	<i>Washington Post</i> , "Melting ice may raise Greenland's value. Trump's fight may be just the start" (Quoted) . [Article]
Dec. 2025	<i>Washington Post</i> , "How Trump's attacks on 'climate alarmism' have already transformed U.S. science" (Quoted) . [Article]
Dec. 2025	<i>Geographical Magazine</i> , "How are scientists and health workers coping with Trump 2.0?" (Interview) . [Article]
Dec. 2025	<i>The Guardian</i> , "Arctic endured year of record heat as climate scientists warn of 'winter being redefined'" (Quoted) . [Article]
Dec. 2025	<i>Science Talk</i> , "Arctic Warming Too Fast Extreme Climate Risks Sea Level Rise Climate Projections beyond 2100?" (Interview) . [Podcast]
Oct. 2025	<i>VernonReporter</i> , "A tool tracking billion dollar disasters is active again after being retired by Trump administration" (Quoted) . [Article]
Oct. 2025	<i>UCI Physical Sciences</i> , "How one Earth system scientist mastered climate science communication" (Interview) . [Article]
Oct. 2025	<i>Carbon Brief</i> , "Antarctic sea ice winter peak in 2025 is third smallest on record" (Quoted) . [Article]
Sep. 2025	<i>Quanta Magazine</i> , "How Climate Scientists Saw the Future Before It Arrived" (Quoted) . [Article]
Aug. 2025	<i>Grist</i> , "Antarctica is in extreme peril" (Quoted) . [Article]
Aug. 2025	<i>CNN</i> , "The strange divide in how Americans experience summer temperatures" (Quoted) . [Article]
Aug. 2025	<i>Harvest Public Media</i> , "The Midwest is getting warmer, even though summer highs are dropping. Here's why" (Quoted) . [Article]
Aug. 2025	<i>Three Degrees Podcast</i> , "Climate Trends, Data, and the Fight Against Disinformation" (Interview) . [Podcast]
Aug. 2025	<i>The Weather Channel</i> , "Phoenix Is Facing A Heat Crisis. Here's A Glimpse of Our Climate Future" (Quoted) . [Article]
Jul. 2025	<i>New York Times</i> , "In a Reversal, Key Hurricane-Monitoring Data Will Stay Online" (Quoted) . [Article]
Jul. 2025	<i>Context</i> , "U.S. hurricane forecasters losing critical access to government data" (Quoted) . [Article]
Jul. 2025	<i>NJ Spotlight News</i> , "Weather forecasting, storm preparedness at risk" (Interview) . [Video]
Jun. 2025	<i>Bloomberg</i> , "This Heat Wave Is Just a Taste of What's to Come" (Quoted) . [Article]
Jun. 2025	<i>Perspectives on Sustainable Development Podcast</i> , "We Live in Systems Not Silos: How Changes in the Arctic Impact the World" (Interview) . [Podcast]
Jun. 2025	<i>Trouw</i> , "This scientist lost his job in the US. Now he's bringing climate change to the public" (Interview) . [Article]
Jun. 2025	<i>ClimateGenn Podcast</i> , "Zack Labe: Science Is being Silenced!" (Interview) . [Podcast]

May 2025	<i>High North News</i> , "US Climate Researcher: "We Are at a Tipping Point"" (Interview) . [Article]
May 2025	<i>StudyFinds</i> , "'Rapid Response System' Can Tell When Heat Waves Are Fueled By Climate Change" (Featured Study) . [Article]
Mar. 2025	<i>Silenced Science Stories</i> , "Dr. Zachary Labe, Climate Scientist" (Interview) . [Article]
Mar. 2025	<i>DW</i> , "Is climate science threatened as US slashes jobs, funding?" (Quoted) . [Article]
Jan. 2025	<i>Yorkshire Bylines</i> , "Time to exit this climate "road to ruin", warns UN secretary general" (Quoted) . [Article]
Oct. 2024	<i>News Decoder</i> , "Extreme weather makes climate change a reality now" (Quoted) . [Article]
Oct. 2024	<i>Carbon Brief</i> , "Antarctic sea ice maximum in 2024 is 'second lowest' on record" (Quoted) . [Article]
Aug. 2024	<i>Carbon Brief</i> , "IPCC meeting in Sofia fails to agree timeline for seventh assessment report" (Quoted) . [Article]
Jul. 2024	<i>Axios</i> , "The seas are not alright: "Unusual warmth" hits nearly all of world's oceans" (Quoted) . [Article]
Jun. 2024	<i>USA Today</i> , "Earth marks 12 straight months of record heat, a bewildering climate change milestone" (Quoted) . [Article]
May 2024	<i>Yorkshire Bylines</i> , "Global levels of CO2 for 2023-24 break all records in human history" (Quoted) . [Article]
Apr. 2024	<i>Climate Quandaries Podcast</i> , "S2E4: Zack Labe" (Interview) . [Podcast]
Mar. 2024	<i>Temple Sustainability Blog</i> , "Making Your Research Matter: Communicating for the Climate" (Quoted) . [Article]
Mar. 2024	<i>Carbon Brief</i> , "Antarctic sea ice 'behaving strangely' as Arctic reaches 'below-average' winter peak" (Quoted) . [Article]
Mar. 2024	<i>Yorkshire Bylines</i> , "The Arctic: The first ice-free summer could arrive in the next decade" (Quoted) . [Article]
Mar. 2024	<i>The Washington Post</i> , <i>Capital Weather Gang</i> , "The groundhog was right: Spring is here ahead of schedule" (Quoted) . [Article]
Feb. 2024	<i>Planet NOAA Podcast</i> , "Welcome to Planet NOAA!" (Interview) . [Podcast]
Nov. 2023	<i>Carbon Brief</i> , "Q&A: Warming of 2°C would trigger 'catastrophic' loss of world's ice, new report says" (Quoted) . [Article]
Oct. 2023	<i>NBC News</i> , "'Gobsmackingly bananas' September of extreme warmth alarms climate scientists" (Quoted) . [Article]
Sep. 2023	<i>Carbon Brief</i> , "'Exceptional' Antarctic melt drives months of record-low global sea ice cover" (Quoted) . [Article]
Sep. 2023	<i>Wired</i> , "The Mysterious 'Warming Hole' in the Middle of the US" (Quoted) . [Article]
Aug. 2023	<i>Wired</i> , "This Brutal Summer in 10 Alarming Maps and Graphs" (Quoted) . [Article]
Jul. 2023	<i>USA Today</i> , "The oceans are unusually hot and on track to get hotter. That's not good." (Quoted) . [Article]
Jun. 2023	<i>CNET</i> , "The Arctic Sea Could Be 'Ice-Free' Within Decades. What That Means for Us" (Quoted) . [Article]
Jun. 2023	<i>Bulletin of the Atomic Scientists</i> , "'Uncharted territory': Warming oceans and disappearing sea ice alarm scientists" (Quoted) . [Article]

May 2023	<i>Wired</i> , "Antarctic Sea Ice Is at Record Lows. Is It an Alarming Shift?" (Quoted). [Article]
May 2023	<i>Carbon Brief</i> , "Montreal Protocol has slowed loss of Arctic sea ice, say scientists" (Quoted). [Article]
Mar. 2023	<i>Carbon Brief</i> , "Arctic sea ice winter peak in 2023 is fifth lowest on record" (Quoted). [Article]
Feb. 2023	<i>WFLA Tampa</i> , "How certain is science that humans are warming the climate?" (Featured Graphics). [Article]
Nov. 2022	<i>Princeton AOS & CIMES Newsletter</i> , "Zack Labe in Conversation" (Interview). [Article]
Nov. 2022	<i>Lawrence Livermore National Laboratory</i> , "New analysis helps reconcile differences between satellites and climate models" (Press Release). [Article]
Nov. 2022	<i>The Daily Beast</i> , "What Happens When Even Scientists Get Doom-Pilled?" (Quoted). [Article]
Sep. 2022	<i>Mongabay</i> , "2022: Another consequential year for the melting Arctic" (Quoted). [Article]
Sep. 2022	<i>Carbon Brief</i> , "Arctic sea ice summer minimum in 2022 is 'joint-10th lowest' on record" (Quoted). [Article]
Sep. 2022	<i>CBS News</i> , "Alarming year for extremes": 2021 saw record-high greenhouse gas, ocean heat and sea levels rise, new report finds" (Quoted). [Article]
Aug. 2022	<i>La Libre</i> , "Cette statistique montre vraiment à quel point l'Arctique a changé, c'est spectaculaire" (Interview). [Article]
Aug. 2022	<i>Financial Times</i> , "Climate graphic of the week: Arctic warming four times faster than rest of the planet, study says" (Quoted). [Article]
Jul. 2022	<i>Forbes</i> , "Yes, It's Getting Hotter - This Simple Graphics Explains" (Quoted). [Article]
Jul. 2022	<i>Politico</i> , "GIF of the Day - POLITICO's Power Switch" (Quoted). [Article]
Jul. 2022	<i>Val 202 - Slovene radio station</i> , "Točka preloma ali točka brez povratka?" (Interview). [Article]
Jul. 2022	<i>Foresight</i> , "Bringing Data Driven Stories to Life with Zack Labe" (Interview). [Article]
Mar. 2022	<i>Gizmodo</i> , "Record-Shattering Heat at Both Poles Is Freaking Scientists Out" (Quoted). [Article]
Mar. 2022	<i>Vice</i> , "'Never Supposed to Happen': North and South Poles See Unprecedented Heat" (Quoted). [Article]
Mar. 2022	<i>Yahoo News</i> , "'Not a good sign': Both of Earth's poles see temperatures 30C and 40C above normal" (Quoted). [Article]
Mar. 2022	<i>Carbon Brief</i> , "Arctic sea ice winter peak in 2022 is 10th lowest on record" (Quoted). [Article]
Jan. 2022	<i>Data Skeptic Podcast</i> , "Explainable climate science" (Interview). [Podcast]
Nov. 2021	<i>EnergieWinde</i> , "Zachary Labes Grafiken zum Klimawandel" (Interview). [Article]
Oct. 2021	<i>Mashable</i> , "Gaping hole opened up in 'Last Ice Area' of the Arctic, NASA images show" (Quoted). [Article]
Oct. 2021	<i>CNN</i> , "Antarctica's last 6 months were the coldest on record" (Quoted). [Article]
Sep. 2021	<i>Blasting News</i> , "From Carlos Nobre to Svein Tveitdal, 6 climate activists share ways to tackle climate change" (Interview). [Article]

Sep. 2021	<i>Carbon Brief</i> , "Arctic sea ice summer minimum in 2021 is '12th lowest' on record" (Quoted). [Article]
Sep. 2021	<i>Arctic Today</i> , "As it approaches its yearly minimum, Arctic sea ice extent its slightly greater than recent ultra-low years" (Quoted). [Article]
Sep. 2021	<i>BreezoMeter</i> , "Top 20 Leading Weather & Climate Voices to Follow in 2021" (Listed). [Article]
Aug. 2021	<i>National Geographic</i> , "Siberia's massive wildfires are unlocking extreme carbon pollution" (Quoted). [Article]
Jul. 2021	<i>The Washington Post</i> , <i>Capital Weather Gang</i> , "Arctic climate change may not be making winter jet stream weird after all" (Featured Study). [Article]
Jul. 2021	<i>Sky News</i> "A crisis of fire and ice: Deadly heatwaves and record ice melt tell a story of extreme weather that's here to stay" (Quoted). [Article]
Jun. 2021	<i>ABC News</i> , "It's Not Too Late: Extreme heat in the Arctic" (Interview). [Article]
Jun. 2021	<i>ABC News</i> , "Heat wave in Russia brings record-breaking temperatures north of Arctic Circle" (Quoted). [Article]
Jun. 2021	<i>The Independent</i> , "Merciless temperatures push Moscow to hottest June day in 142 years, as heatwave hits Russia" (Quoted). [Article]
May 2021	<i>Science Magazine</i> , "Landmark study casts doubt on controversial theory linking melting Arctic to severe winter weather" (Featured Study). [Article]
Apr. 2021	<i>Inverse</i> , "What will the Earth look like in 2121? 10 young leaders give their predictions" (Interview). [Article]
Mar. 2021	<i>Carbon Brief</i> , "Arctic sea ice winter peak in 2021 is joint-seventh lowest on record" (Quoted). [Article]
Mar. 2021	<i>Bloomberg</i> , "Arctic Lightning Strikes Tripled in a Decade" (Quoted). [Article]
Feb. 2021	<i>KCSU FM Radio</i> , "Explaining the polar vortex, U.S. COVID-19 deaths pass 500,000" (Interview). [Radio]
Feb. 2021	<i>Weather Geeks Podcast</i> , "Arctic Report Card 2020" (Interview). [Podcast]
Jan. 2021	<i>Blasting News</i> , "Zack Labe on Arctic climate change: It's important to communicate data in an accessible way" (Interview). [Article]
Jan. 2021	<i>CBS News</i> , "Sudden stratospheric warming could mean wild winter storms ahead" (Quoted). [Article]
Jan. 2021	<i>Inside Climate News</i> , "Many Scientists Now Say Global Warming Could Stop Relatively Quickly After Emissions Go to Zero" (Quoted). [Article]
Dec. 2020	<i>Australian Broadcasting Corporation</i> , "The vanishing Arctic" (Quoted). [Article]
Dec. 2020	<i>Cambridge Climate Lecture Series: Shaping the Future Podcast</i> , "Decoding NOAA's Arctic Report Card With Contributing Climate Scientist, Dr Zack Labe" (Interview). [Podcast]
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PERSONAL INTERESTS

Vegetable gardening, travel, cooking, hiking, fishing, lighthouses, trains, winter synoptic storms, long-range and seasonal weather forecasting